Yujie Zheng
Department of Earth Sustainable Earth Systems Sciences
School of Natural Science and Mathematics
University of Texas at Dallas

Educational history:

Ph.D., 2020, Stanford University, Stanford, CA, USA, Geophysics
Bachelor of Science, 2014, Peking University, Beijing, China, Geophysics

Employment history

Assistant Professor, 2023 - present, University of Texas at Dallas, Richardson, TX, USA
Postdoctoral Scholar, 2020 – 2023, California Institute of Technology, Pasadena, CA, USA
Graduate Research Assistant, 2014 – 2020, Stanford University, Stanford, CA, USA

Professional recognitions and honors

Outstanding Student Paper Award, 2017, American Geophysics Union
The Joshua L. Soske Fellowship, 2014, Stanford University
Principle's Award for undergraduate research, 2012, Peking University
Chinese Universities Study Award, 2012, National University of Singapore
The May 4th Fellowship, 2010, Peking University

Mentoring

Honors and awards of mentees

Xiange Jian, western groundwater congress scholarship, 2026
Eshanta Mishra, Earthscope career internship, 2026

Mentees

Rishabh Dutta, Postdoc in Earth Sciences, UTD	2025-now
Xiange Jian, Ph.D. candidate in Earth Sciences, UTD	2024-now
Eshanta Mishra, Ph.D. candidate in Earth Sciences, UTD	2024-now
Mauritz Feldbrugge, Ph.D. student in Earth Sciences, UTD	2026-now
Lily Chhoeung, undergrad in Statistics, UTD	2026-now
Ananya Sankaran, undergrad in Data Sciences, UTD	2025-now
Jesus Martinez, undergrad in Physics, UTD	2024
Renny Hoang, undergrad in Computer Sciences, UTD	2024
Hailey Whitley, undergrad in Physics, UTD	2023

Teaching

GEOS4395 Remote Sensing Geophysics, UTD	2024, 2025, 2026
GEOS5328 Imaging Radar Principles and Geophysical Applications, UTD	2025, 2026

Research Grants (~ $1.4 *million* as PI)

1. **Cascadia Region Earthquake Science Center seed grant program (PI),** Precise Localization and Measurement of Slow Slip Events in Central Cascadia with InSAR, 2025 - 2026

2. **NASA Earth Surface and Interior Program (PI),** Four-Dimensional Characterization of Groundwater Recharge from Atmospheric Rivers: A combined Geodetic and Seismic Analysis in Greater Los Angeles, 2025 - 2028
3. **NASA NISAR DART Team (PI),** Multi-Frequency NISAR Closure Phase and Radar Observables for Monitoring Managed Water Stress in Water-Limited Agriculture, 2027-2029 (selected)

Publications

*Graduate student mentee

Journal Papers

[13] Jones, C.E., Rosenqvist, A., Rommen, B., Fitrzyk, M., Rignot, E., Scheuchl, B., **Zheng, Y.,** Hooper, A., Simons, M., Kobayashi, T., Siqueira, P., Santoro, M., Tadono, T., Hess, L., McNairn, H., Frulla, L., Johansson, M., Dierking, W., Rouyet, L., Bartsch, A., Lopez Dekker, P., Johannessen, J.,(2026), Observation and coordination needs for current, near-future, and next generation Earth-observing SAR systems, ***Earth and Space Science***, https://doi.org/10.1029/2025EA004868

[12] **Zheng, Y.,** Fattahi, H., (2026) Modeling, Prediction and Retrieval of Surface Soil Moisture from InSAR Closure Phase, ***Remote Sensing of Environment***, 333, 115104. https://doi.org/10.1016/j.rse.2025.115104

[11] **Zheng, Y.,** Simons, M, (2025), Integrating long-term InSAR monitoring into local groundwater management: Insights from the Hollywood Basin, California, ***Water Resources Research***, 61(11), e2024WR039161. https://doi.org/10.1029/2024WR039161

[10] Tay, C., Jovani-Sancho, A., Yulianti,L., Evans, C., Callaghan, N., Jaya A., Salman, R., **Zheng, Y.,** Susilo, S., Dohong, S., Yun, S. (2025), Satellite radar advances carbon emissions accountability over tropical peat, ***Communications Earth & Environment***, 6(1), 971. https://doi.org/10.1038/s43247-025-02926-6

[9] Zhang, S., Luo, B., **Zheng, Y**., Lumley D., Sen, M., Zhu, H., (2025), Mapping spatiotemporal variations of near-surface seismic velocity to monitor groundwater in central Oklahoma, ***Geophysical Research Letters***, https://doi.org/10.1029/2025GL115201

[8] Mao, S., Ellsworth, W., **Zheng, Y.,** Beroza, G., (2025), Depth-Dependent Seismic Sensing of Groundwater Recovery from the Atmospheric-River Storms of 2023, ***Science***, https://doi.org/adr6139

[7] **Zheng, Y.,** Fattahi, H., Agram, P., Simons, M., and Rosen, P., (2022), On closure phase and Systematic Bias in Multi-looked SAR Interferometry. ***IEEE Transactions on Geoscience and Remote Sensing***, http://doi.org/10.1109/TGRS.2022.3167648

[6] **Zheng, Y.,** Laura Blackstone and Segall, P., (2022), Constraints on absolute magma chamber volume from geodetic measurements of Trapdoor faulting in the Galapagos. ***Geophysical Research Letters***, https://doi.org/10.1029/2021GL095683

[5] Wang, T., **Zheng, Y.**, Pulvirenti, F., Segall, P., (2021). Post-2018 caldera collapse reinflation uniquely constrain Kilauea's magmatic system, ***Journal of Geophysical Research: Solid Earth***, https://doi.org/10.1029/2021JB021803

[4] **Zheng, Y.**, Zebker, H.A., and Michaeledes, R.J., (2021). A New Decorrelation Phase Covariance Model for Noise Reduction in Unwrapped Interferometric Phase Stacks, ***IEEE Transactions on Geoscience and Remote Sensing***, https://doi.org/10.1109/TGRS.2021.3050087

[3] Michaelides, R.J., Zebker, H.A., **Zheng, Y.**, (2019). An Algorithm for Estimating and Correcting Decorrelation Phase from InSAR Data Using Closure Phase Triplets. ***IEEE Transactions on Geoscience and Remote Sensing***, http://doi.org/10.1109/TGRS.2019.2934362

[2] **Zheng, Y.** and Zebker, H.A., (2017). Phase Correction of Single-Look Complex Radar Images for User-Friendly Efficient Interferogram Formation. ***IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing***, 10(6), http://doi.org/10.1109/JSTARS.2017.2697861

[1] **Zheng, Y**. and Zhou, S., (2014). The spatiotemporal variation of the b-value and its tectonic implications in North China. ***Earthquake Science***, https://doi.org/10.1007/s11589-014-0086-8

Refereed conference publications:

[3] Mishra, E.*, **Zheng, Y**., & Jian, X.* (2026). Monitoring Slow Slip Events in Central Cascadia using InSAR. *2026 IEEE International Geoscience and Remote Sensing Symposium,* http://doi.org/10.1109/IGARSS60577.2026.11730108

[2] **Zheng, Y.,** Zebker, H.A., and Michaelides, R.J., (2020) "A Physics-Based Decorrelation Phase Covariance Model for Effective Decorrelation Noise Reduction in Interferogram Stacks," *2020 IEEE International Geoscience and Remote Sensing Symposium*, http://doi.org/10.1109/IGARSS39084.2020.9323237

[1] Zebker, H.A. and **Zheng, Y**. (2016), Robust and efficient InSAR deformation time series processing, *2016 IEEE International Geoscience and Remote Sensing Symposium*, http://doi.org/10.1109/IGARSS.2016.7729827

Works in progress:

[2] Mishra, E.*, Jian X.*, and Zheng, Y., in preparation, Monitoring slow slip events in Central Cascadia using InSAR

[1] Jian, X.*, Mishra E.*, and Zheng, Y in preparation, Quantifying groundwater storage changes in California's San Gabriel Basin from 1992 to 2026 using multi-mission InSAR

Invited talks/presentations at professional meetings, seminars

2026 John D. Pigott Colloquium Series, University of Oklahoma, Norman, OK, October 8
2026 SEG/AGU Hydrogeophysics workshop, Boise, Idaho, July 20-22
2026 COMET+ Webinar Series, University of Leeds, UK, remote, February 26.
2025 Department of Earth Sciences, University of Texas, Dallas, September 4.
2024 NASA Solid Earth Team "Core 2.0" workshop, Washington, DC, September 16-18
2024 SMU Energy, Environment and Natural Resources Colloquium, Dallas, TX, May 2.
2023 Geoclub Seminar, University of Texas, Dallas, November 10
2023 Roy Huffington Department of Earth Sciences, South Methodist University, October 6.
2022 Department of Earth Sciences, University of Texas, Dallas, November 3.
2022 NISAR Community Science Workshop, Pasadena, CA, August 17.
2022 School of Earth Sciences, Ohio State University, April 13.
2022 Department of Earth Sciences, University of Hong Kong, April 3.
2022 Department of Earth and Atmospheric Sciences, Indiana University, January 26.
2021 Kimbell School of Geosciences, Midwestern State University, December 6.

2021 SCEC Community Geodetic Model Workshop, remote, November 29.
2020 Seismological Laboratory, California Institute of Technology, November 20.
2018 Annual AGU Fall meeting, San Francisco, CA, Dec 15-19

Presentations at professional meetings (first author only)
Aug 2026 Zheng, Y., Simons, M., Integrating long-term InSAR monitoring into local groundwater management: Insights from the Greater Los Angeles, California. IEEE IGARSS
Dec 2025 Zheng, Y., Fattahi, H., Modeling, Prediction, and Retrieval of Surface Soil Moisture from InSAR Closure Phase. AGU Fall Meeting
Dec 2025 Zheng, Y., Simons, M., Integrating long-term InSAR monitoring into local groundwater management: Insights from the Hollywood Basin, California. AGU Fall meeting
Aug 2025 Zheng, Y., Fattahi, H., Modeling, Prediction, and Retrieval of Surface Soil Moisture from InSAR Closure Phase. IEEE IGARSS
Dec 2023 Zheng, Y., Simons, M., Ups and Downs of Beverly Hills, California – Analysis of Surface Deformation Derived from InSAR: 1992-2023. AGU Fall Meeting
Sep 2023 Zheng, Y., Fattahi, H., Modeling soil moisture with cumulated closure phase of interferometric SAR measurements. Fringe 2023.
Mar 2023 Zheng, Y., Simons, M. Ups and Downs of Beverly Hills, California: Analysis of surface deformation from InSAR. GAGE/SAGE 2023 Community Science Workshop
Dec 2022 Zheng, Y., Fattahi, H., Modeling soil moisture with cumulated closure phase of interferometric SAR measurements. AGU Fall Meeting.
Dec 2021 Zheng, Y., Fattahi, H., Agram, P., Simons, M., Rosen, P., On Closure phase and Systematic Bias in Multi-looked SAR Interferometry. AGU Fall Meeting.
Dec 2021 Zheng, Y., Simons, M., Investigating land surface displacements over the San Gabriel Valley, California. AGU Fall Meeting.
Jun 2021 Zheng, Y., Fattahi, H., Agram, P., Simons, M., On Closure phase and Systematic Bias in Multi-looked SAR Interferometry. Fringe Workshop.
Dec 2020 Zheng, Y., Fattahi, H., Agram, P., Simons, M., Assessing closure phase and its impact on InSAR time-series. AGU Fall Meeting.
Sep 2020 Zheng, Y., Zebker, H.A. and Michaelides, RJ.,. A Physics-Based Decorrelation Phase Covariance Model for Effective Decorrelation Noise Reduction in Interferogram Stacks. IEEE International Geoscience and Remote Sensing Symposium.
Dec 2019 Zheng, Y. and Zebker, H.A., Are redundant interferograms really redundant? On the use of redundant interferograms to reduce noise. AGU Fall Meeting.
Dec 2018 Zheng, Y. and Zebker, H.A., Slow Slip Events in Cascadia: Observation from Sentinel-1. AGU Fall Meeting.
Dec 2017 Zheng, Y., and Zebker, H.A., Retrieving Ground Deformation Associated with Cascadia Slow Slip Events Using Sentinel-1 Data. AGU Fall Meeting.
Dec 2016 Zheng, Y. and Zebker, H.A., Crustal deformation associated with Cascadia slow slip events from InSAR time-series, AGU Fall Meeting
Jul 2014 Zheng, Y. and Zhou, S., The spatiotemporal variation of the b-value and its tectonic implications in North China, International Workshop on Statistical Seismology

Professional memberships:
Member, American Geophysical Union, 2014 – present
Member, Institute of Electrical and Electronics Engineers, 2025 -- present

Service

University/School/Department Services

- Member, SESS Department Head Search Committee, 2026-2027
- Member, SESS Hydrology Search Committee, 2025-2026
- Member, SESS scholarship committee, 2026
- SESS Undergraduate Advisor, 2024-2026
- Member, SESS Seminar Committee, 2023-2024
- Member, NSM strategy committee, 2023-2024
- Member, NSM teaching workload committee, 2023

Professional Service

- Journal reviewer
  Nature Communications; Remote Sensing of Environment; Water Resources Research; IEEE Transactions on Geoscience and Remote Sensing; IEEE Transactions on Parallel and Distributed Systems; Journal of Geodesy; IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing; Computers & Geosciences; Earth and Space Science; IEEE Geoscience and Remote Sensing Letters; Remote Sensing Applications: Society and Environment; Remote Sensing in Earth System Science.
- Grant reviewer
  NASA Experimental Program to Stimulate Competitive Research, ad hoc reviewer, 2017
  NASA FINESST Program, panelist, June 4th to June 6th, 2024
- Conferences
  Session Chair, Methodological Aspects in InSAR/DInSAR Analysis , IGARSS (2025)
  AGU Geodesy Section OSPA Committee (2024)
  Primary Convenor and Chair, Recent Advances in SAR/InSAR, AGU (2024 – present)